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# The Impact of Structured ESG KPI Presentation Methods on Stakeholder Engagement: An LLM-Based Persona Simulation Study

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## Abstract

Environmental, Social, and Governance (ESG) information disclosure has become a core element of corporate transparency. However, small and medium-sized enterprises (SMEs) and social enterprises face challenges in designing effective ESG communication strategies due to resource constraints. This study empirically analyzes how ESG information presentation methods affect stakeholder evaluation and participation decisions using large language model (LLM)-based persona simulation.

The experiment was conducted on a hypothetical Company K constructed from real corporate ESG disclosure data, comparing three conditions that differed only in information presentation methods. Condition A provided narrative explanations only, Condition B presented quantitative indicators and quarterly dashboards following the 3-Tier ESG KPI framework, and Condition C presented only quantitative KPIs in structured table format. A total of 120 diverse stakeholder personas evaluated the company over 12 quarters using 5 different LLM models, generating 21,600 evaluation data points.

Analysis revealed significant differences in participation rates and evaluation scores depending on the information presentation method. In particular, Condition B, which combined KPIs with visual dashboards, showed the highest participation rate compared to Condition C (simple KPI) and Condition A (narrative), indicating that participation decisions were most strongly reinforced when information structuring and visualization were combined for the same company. However, differences in evaluation scores were relatively limited compared to participation rates.

This study empirically examines the role of information structuring and visualization by applying cognitive load theory and dual coding theory to the ESG communication context. It also suggests that LLM-based persona simulation can serve as a methodological alternative enabling large-scale experiments in ESG communication research.

# **1. Introduction**

## **1.1 Research Background**

With the global spread of Environmental, Social, and Governance (ESG) management, accurately and transparently measuring and reporting corporate ESG performance has emerged as a key challenge for securing corporate transparency and credibility [1]. However, SMEs and social enterprises face structural difficulties in establishing complex Key Performance Indicator (KPI) systems and effectively communicating them to stakeholders due to limited resources and expertise required for ESG information disclosure [2].

Existing ESG communication research has mainly focused on sustainability reports and disclosure strategies of large corporations, with relatively limited empirical analysis of how differences in information presentation methods affect stakeholder evaluation and participation decisions [3]. In particular, systematic research on what structure and format should be used to present multidimensional and complex performance information like ESG to enhance stakeholder understanding and participation has not been sufficiently accumulated.

Meanwhile, recent developments in large language models (LLMs) present new methodological possibilities to address this research gap. Researchers at Stanford HAI [4] showed that LLM simulations based on 1,052 real individuals could reproduce General Social Survey responses with approximately 85% accuracy. This suggests that it is possible to simulate the decision-making processes of diverse stakeholder personas on a large scale while mitigating the time, cost, and ethical constraints of traditional surveys and experiments.

In this context, LLM-based persona simulation can be utilized as a useful alternative methodology to systematically verify the effects of information presentation methods in ESG communication research, where large-scale participant recruitment is challenging.

## **1.2 Research Purpose**

This study aims to analyze how differences in ESG campaign information presentation methods affect stakeholder participation decisions and evaluations using LLM-based persona simulation. Specifically, this study compares how stakeholder responses differ according to the level of information structuring and the presence of visualization elements, centered on the 3-Tier ESG KPI framework. For this purpose, ESG information presentation methods are divided into three conditions for analysis: Condition A presents ESG activities in narrative text only, Condition B provides quantitative indicators and quarterly dashboards based on the 3-Tier KPI system, and Condition C presents only quantitative indicators in structured format without visualization elements, using the same KPI system.

## **1.3 Research Questions**

This study seeks to answer the following research questions:

RQ1: What differences appear in stakeholder participation intentions and evaluation scores according to differences in ESG information presentation methods?

RQ2: How do the effects of information presentation methods differ according to stakeholders' demographic characteristics and information processing tendencies?

RQ3: What patterns of change do the effects of information presentation methods show over time in repeated evaluation processes?

RQ4: To what extent do simulation results using different LLM models show consistency?

## 1.4 Research Hypotheses

Based on cognitive load theory and dual coding theory, this study establishes the following hypotheses:

H1 (Main Hypothesis): Information presentation method combining structured KPI and visualization elements (Condition B) will induce higher participation intentions and evaluation scores than the method presenting only quantitative KPIs (Condition C), and Condition C will show superior results to narrative information presentation method (Condition A).

H2 (Moderating Hypothesis): The effects of information presentation methods will be moderated by stakeholder characteristics such as technology acceptance, information processing tendencies, and environmental concern.

## 2. Related Research

### 2.1 KPI-Based ESG Campaign Indicator Design Research

Prior indicator design research for quantifying ESG campaign performance has been discussed in the areas of Environment (E), Social (S), and Governance and Value (G), centered on global guidelines and national standards [5-7]. Key evaluation indicators and their research basis for each area are as follows.

Carbon reduction contribution in the environmental area is primarily based on the waste incineration emission calculation logic presented by IPCC 2006 guidelines [5]. In particular, the 2024 National Greenhouse Gas Inventory Report defines material-specific emission factors such as PET (2.29kg CO<sub>2</sub>/kg) and HDPE (3.12kg CO<sub>2</sub>/kg) reflecting domestic plastic composition [6]. Furthermore, the ESG Best Practice Code recommends applying differential correction factors for reuse and upcycling according to resource circulation principles, and CDP guidelines present standards for calculating environmental benefits through the concept of avoided emissions from a product lifecycle perspective [7]. Meanwhile, energy savings compared to new production (PET 13.9kWh/kg, HDPE 12.5kWh/kg, etc.) are also utilized as measurement indicators through product Life Cycle Assessment (LCA) data [13].

Community impact in the social area has been systematized through GRI 413 and the Social Value Index (SVI) model [8], [9]. Employment creation value is calculated by combining number of employees, working hours, and wages according to relevant manuals presenting evaluation methods based on actual personnel and performance indicators [10]. For educational campaigns, GRI 413-1 guidelines suggest measuring the qualitative reach of education by differentiating simple awareness-level promotion from direct education involving actual participation according to weighting [8]. This indicator design is contextually aligned with the MSCI Human Capital Development Indicator requirement for substantial capacity building and growth management systems for human resources [11].

Economic performance of the circular economy is designed based on Ellen MacArthur Foundation's value preservation theory [13]. Korea Environment Corporation provides basic data for calculating resource value preservation amounts through actual transaction price data [12], and measures decoupling performance between resource consumption and growth through the upcycling value-added rate, which excludes raw material value from output value according to UN System of National Accounts (SNA) production account logic [14]. This serves as a key indicator demonstrating the linkage between economic value creation and environmental resource preservation.

These ESG performance indicator design studies have developed focusing on calculation validity and international consistency of individual indicators, but analysis of how these indicators substantively affect participation decisions when presented to stakeholders is relatively lacking. This study focuses on stakeholder response differences when the same KPI system is communicated through different information presentation methods, based on these indicator design discussions.

## **2.2 Cognitive Structure-Based Learning Design Theory and LLM-Based Behavioral Simulation**

Cognitive Load Theory, developed in the late 1980s, assumes that working memory capacity is limited to  $7 \pm 2$  chunks. Three types of cognitive load occur depending on information presentation methods: intrinsic load (inherent complexity of the task), extraneous load (unnecessary load due to presentation method), and germane load (load for schema construction). Simple narrative (Condition A) induces high extraneous load due to unstructured information, causing excessive consumption of cognitive resources in information search and integration. In contrast, structured KPIs (Conditions B and C) minimize extraneous load through clear hierarchical structure and consistent format, enabling stakeholders to focus cognitive resources on germane load (understanding the meaning of ESG performance).

According to Allan Paivio's Dual Coding Theory, the human mind processes information through two independent but interconnected channels: verbal channel (logogens) and nonverbal/image channel (imagens). Information encoded simultaneously in both systems shows better retention and retrieval than a single system. Dashboards (Condition B) provide dual coding by combining numerical data (verbal channel) with graphs and color coding (visual channel), while narratives (Condition A) mainly utilize only the verbal channel [19]. Multimedia learning research suggests that visual information is often processed more efficiently than text. Therefore, Condition B is expected to enable more efficient information processing and memory than Condition C.

Recent advances in LLM capabilities have enabled sophisticated persona simulation for social science research. LLM-based persona simulation is a research methodology that simulates human responses by assigning specific demographic and psychological characteristics to large language models. Argyle et al. [4] demonstrated in their "Out of One, Many" study published in Political Analysis that diverse human samples could be simulated from a single LLM, while Hu and Collier [21] quantitatively confirmed at ACL that persona variables significantly explain LLM response variance. In particular, Harvard Business School's Brand, Israeli, and Ngwe research team [22] confirmed that LLM simulation results showed high similarity to 9,300 human responses in 57 consumer product surveys. Their developed Semantic Similarity Rating (SSR) method achieved 90% of human test-retest reliability by mapping LLM text responses to Likert scale distributions.

The synergistic relationship between Cognitive Load Theory and Dual Coding Theory explains why Condition B (KPI+Dashboard) demonstrates superior effectiveness. These two theories operate through complementary mechanisms: CLT addresses how information structure affects cognitive resource allocation, while DCT explains how multi-modal encoding enhances retention and retrieval.

Mayer's Cognitive Theory of Multimedia Learning (CTML) [24] provides an integrative framework through three core principles. First, the dual-channel assumption states that visual and verbal information are processed through separate but interconnected pathways. Condition B activates both channels—numerical KPIs engage verbal-symbolic processing while trend graphs engage visual-spatial processing. Second, the limited-capacity assumption recognizes that each channel has finite working memory ( $\sim 7 \pm 2$  chunks for verbal,  $\sim 3$ -5 objects for visual). Condition

A's narrative imposes high extraneous load by requiring readers to mentally extract and organize scattered ESG information. Condition B reduces extraneous load through hierarchical structure (3-Tier framework) and spatial organization (dashboard layout). Third, the active processing assumption states that learning requires selecting, organizing, and integrating information. Dashboards facilitate all three: color-coding guides attention (selection), hierarchical tiers provide structure (organization), and trend lines enable pattern recognition (integration).

This integrated framework predicts that Condition B enables both rapid pattern recognition (System 1: "green upward trends = improving performance") and deliberate verification (System 2: examining specific KPI values), while Condition A relies primarily on effortful System 2 processing and Condition C provides System 2 data without System 1 visual affordances. The additive effect observed in Table 4 (structuring: +13.87%p, visualization: +15.17%p, combined: +29.04%p) supports this dual-pathway explanation.

### 3. Proposed Method

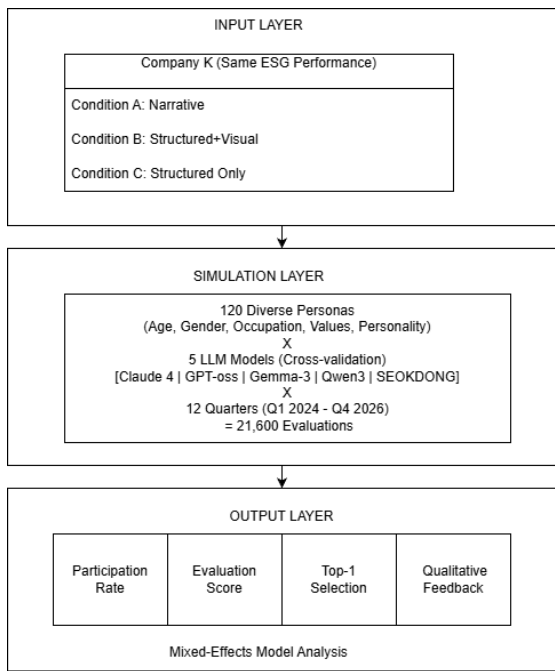


Figure 1. Method Summary

This study proposes a large language model (LLM)-based persona simulation methodology to verify the impact of ESG information structuring levels on stakeholder campaign participation decisions. While existing ESG disclosure research has shown limitations in sample size and diversity by relying on actual stakeholder surveys or experimental participant recruitment, this study aimed to secure statistical stability for analysis and observe diverse response patterns through large-scale simulation of 21,600 cases using virtual personas assigned various demographic and psychological characteristics to LLMs.

The core hypothesis of the proposed study is "when ESG information is presented in a hierarchically structured manner, stakeholder participation intentions and trust are improved

compared to narrative information presentation." The analytical framework of this study is theoretically based on Information Processing Theory and Cognitive Load Theory.

According to Kahneman [18,19], as information complexity increases, attention resource consumption increases, and structured formats reduce cognitive load to improve decision-making quality. Furthermore, Fehrenbacher and Soderstrom's [20] eye-tracking study empirically confirmed that ESG information requires longer gaze fixation times than financial information, and additional cognitive effort is needed particularly for understanding ESG information labels. Since ESG information simultaneously includes multidimensional elements of environment, social, and governance, it is highly likely to impose a high cognitive burden on stakeholders when presented in an unstructured format.

### 3.1 3-Tier ESG KPI Framework

The 3-Tier ESG KPI framework proposed in this study was designed considering consistency with domestic and international ESG evaluation standards such as IPCC, GRI, CDP, and KCGS. This system is a three-tiered hierarchical structure centered on maximizing environmental (E) circularity, providing a framework for systematically measuring complex ESG performance and effectively communicating it to stakeholders.

*Table 1. 3-Tier ESG KPI Framework Structure*

| Tier   | Category                       | Specific Indicators                                                                                                                                                                                                   |
|--------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1 | Core Performance Indicators    | <ul style="list-style-type: none"> <li>• Carbon reduction (CO<sub>2</sub> tons)</li> <li>• Waste conversion rate (%)</li> <li>• Social value creation (elderly jobs, environmental education participants)</li> </ul> |
| Tier 2 | Supporting Activity Indicators | <ul style="list-style-type: none"> <li>• Energy savings (kWh)</li> <li>• Number of partner organizations</li> <li>• Collection and processing volume (units)</li> </ul>                                               |
| Tier 3 | Integrated Impact Indicators   | <ul style="list-style-type: none"> <li>• Integrated ESG Impact Score (out of 100)<br/>- Environment (50%) + Social (30%) + Governance (20%)</li> </ul>                                                                |

#### 3.1.1 Tier 1: Core Performance Indicators

Tier 1 consists of top-level indicators that most intuitively represent the organization's ESG impact. Measured quarterly, they are designed to enable stakeholders to immediately grasp the organization's core performance. Carbon reduction measures the CO<sub>2</sub> emissions reduced through toy recycling in tons, waste conversion rate represents the percentage of waste reused or recycled instead of landfilled or incinerated, and social value creation is quantified by the number of elderly jobs created and participants in environmental education programs.

#### 3.1.2 Tier 2: Supporting Activity Indicators

Tier 2 consists of supporting indicators that explain how Tier 1's core performance was achieved. Energy savings measure the electricity saved through recycling processes in kWh, number of partner organizations indicates the degree of community partnership and network expansion, and collection and processing volume tallies the number of toys collected and processed quarterly. These indicators provide context and process for Tier 1 performance, enabling stakeholders to understand the organization's activities more deeply.

### 3.1.3 Tier 3: Integrated Impact Indicators

Tier 3 is a final indicator that integrates Tier 1 and Tier 2 performance into a single comprehensive score. Calculated annually, it is standardized on a 100-point scale with weights of 50%, 30%, and 20% applied to the three axes of Environment (E), Social (S), and Governance (G) respectively. This integrated score enables at-a-glance comparison of the organization's overall ESG performance and allows for year-over-year trend analysis and benchmarking. The weights were set to prioritize environmental circularity while also reflecting social and governance aspects in a balanced manner.

## 3.2 LLM-Based Persona Simulation Evaluation Framework

### 3.2.1 Persona Injection Methodology

This study injected personas into LLMs using role prompting techniques. When designing prompts, the following elements were explicitly assigned to each persona, referring to the PersonaLLM framework [23] recommendations:

"You are a [age]-year-old [gender] who has been working in [occupation] for [years of experience] years. Residential area: [region], Education level: [education], Core values: [key value 1], [key value 2], [key value 3], Investment experience: [investment experience level], Interest in ESG: [interest level]. As a person with the above characteristics, please review the following ESG campaign information and determine whether you are willing to participate in the campaign."

The Big Five personality model was applied to design the psychological characteristics of personas. Personality diversity of personas was secured by combining high/low levels across five dimensions: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness.

### 3.2.2 Evaluation Condition Design (Conditions A/B/C)

This study designed three experimental conditions to verify stakeholder response differences according to ESG information presentation methods:

*Table 2. Characteristics of ESG Information Presentation Conditions*

| Condition   | Presentation Method | Characteristics                                                    |
|-------------|---------------------|--------------------------------------------------------------------|
| Condition A | Narrative           | Company narrative-centered, no quantitative indicators             |
| Condition B | KPI + Dashboard     | Quantitative KPI values, visual information provided via dashboard |
| Condition C | KPI Only            | Only quantitative KPI values provided                              |

Condition A (narrative information presentation) describes the company's ESG activities in explanatory sentences without quantitative indicators. For example, it presents goals and activities in a narrative-centered manner such as "Our company is pursuing various environmental initiatives to achieve carbon neutrality by 2030." This reflects the traditional disclosure method commonly used in existing ESG reports and focuses on qualitatively conveying the company's intentions and direction.

Condition B (KPI + dashboard information presentation) presents key KPIs quantitatively for each area of Environment, Social, and Governance according to the 3-Tier ESG KPI framework proposed in this study, and provides them visualized in quarterly dashboard format. Each KPI includes current values, year-over-year change rates, and levels compared to industry averages to ensure comparability. The dashboard consists of the following elements: (1) Time-series trend graphs showing changes in quarterly core indicators (Tier 1), (2) Progress bars visualizing current achievement levels against annual targets, (3) Comparison charts showing increase/decrease rates compared to the previous quarter.

Condition C (simple KPI presentation) is based on the same 3-Tier ESG KPI system as Condition B, but presents only quantitative KPI values in table format without dashboards or additional visualization elements. This was designed to maintain the information structuring effect while isolating and analyzing the impact of visual representation elements on participation decision-making.

### **3.2.3 Evaluation Items and Measurement Indicators**

When requesting LLM personas to evaluate each ESG campaign, the following four items were measured:

1. Participation intention: "Are you willing to participate in this ESG campaign?" (Yes/No)
2. Participation score: "Please rate your intention to participate in the campaign on a 1-10 scale."
3. Reason description: "Please explain in detail why you made the above judgment."
4. Top campaign selection: "Among the 5 campaigns presented, select the one you most want to participate in."

In this study, rather than directly measuring trust as a separate scale, it was indirectly inferred through comprehensive analysis of participation intentions, participation scores, selection patterns, and qualitative responses.

## **4. Research Method**

This study designed a simulation experiment using hypothetical ESG companies to analyze the impact of ESG information presentation methods on stakeholder participation choices and evaluation behaviors. The experiment adopted a mixed-factor design of 3 (information presentation conditions)  $\times$  120 (personas)  $\times$  5 (LLM models)  $\times$  12 (quarters), generating a total of 21,600 evaluation data points.

The experiment included five hypothetical ESG companies, designated as Companies K, L, M, N, and O. All companies were set to perform certain levels of ESG activities in environmental, social, and governance areas, and participants were asked to evaluate all five companies simultaneously each quarter and choose whether to participate.

Among these, Company K was set as the core analysis target of this study. Company K was presented differently by experimental condition, maintaining the same ESG performance level and business content while varying only the information presentation method. In contrast, Companies L, M, N, and O maintained the same information structure and content across all experimental conditions, serving as control groups. This design controlled for observed differences in participation choices and evaluation scores to be attributable to differences in information presentation methods rather than inherent differences between companies.



The information presentation methods applied to Company K are divided into three conditions. Condition A explains ESG activities in qualitative narrative form without including quantitative performance indicators. Condition B quantitatively presents key indicators based on the 3-Tier ESG KPI framework and provides visual information through quarterly dashboards. Condition C uses the same KPI system as Condition B but presents only quantitative indicators in structured table format without visual dashboards.

Participants were allowed to assign scores to all five companies each quarter and select multiple companies they were willing to participate in. This study was designed to observe differences in participation selection rates and evaluation scores when only the information presentation method changed for the same Company K through this repeated evaluation structure. The purpose is to isolate and analyze the effect of information structure itself rather than differences in performance between companies.

#### **4.1 Persona Development**

This study developed a total of 120 personas to reflect diverse stakeholder perceptions and decision-making characteristics. Personas were constructed comprehensively considering demographic dimensions such as age, gender, education level, occupation, and region, as well as psychological dimensions including personality traits, values, lifestyle, and decision-making tendencies. Personas were stratified sampled based on major demographic variables to secure representativeness, and background information, value systems, and behavioral tendencies were explicitly defined for each persona. This ensured that consistency and differentiation of responses between personas were maintained during the LLM-based simulation process.

#### **4.2 LLM Models and Parameters**

Five large language models were utilized for cross-model validation. The models used were Claude 4 Sonnet, GPT-oss-20B, Gemma-3-12B, Qwen3-30B, and SEOKDONG-llama3.1. This multi-model approach was adopted to verify consistency of results across models with different architectures and learning characteristics. All models were executed under identical parameter conditions, with temperature value fixed at 0.7 to maintain balance between generation consistency and diversity.

##### **4.2.1 Computational Resources and Infrastructure**

All experiments were conducted on a single workstation with the following specifications:

- GPU: NVIDIA GeForce RTX 3090 24GB
- CPU: AMD Ryzen 5 7600 (6-core, 12-thread, 3.8GHz base clock)
- RAM: 64GB DDR5 (dual-channel)
- Storage: 1TB NVMe SSD

Model execution approach varied by accessibility:

- Claude 4 Sonnet: Accessed via commercial API endpoints
- GPT-oss-20B, Gemma-3-12B, Qwen3-30B, SEOKDONG-llama3.1: Local inference using HuggingFace Transformers library with 4-bit quantization (bitsandbytes) to fit within 24GB VRAM constraints

Data collection timeline: December 15-30, 2025 (16 days total)

Average inference time per evaluation (single persona, single quarter, single model):

- Claude 4 Sonnet (API): 1.6975 seconds
- GPT-oss-20B (local): 1.0309 seconds
- Gemma-3-12B (local): 3.3751 seconds
- Qwen3-30B (local): 4.9772 seconds
- SEOKDONG-llama3.1 (local): 1.0342 seconds

Total computational load: 21,600 evaluations ( $120 \text{ personas} \times 5 \text{ models} \times 12 \text{ quarters} \times 3 \text{ runs}$  for stochasticity management) resulted in approximately 64,800 total LLM inferences. Estimated total GPU compute time: ~52 hours for local models, with API calls processed concurrently.

Batch processing strategy: Evaluations were parallelized by model and quarter. For local models, batch size was set to 8 evaluations per forward pass to optimize GPU utilization while maintaining memory constraints. API calls were rate-limited according to provider restrictions (Claude: 50 requests/minute)

Software environment:

- Python 3.13.3
- PyTorch 2.8.0
- Transformers 4.35.2
- CUDA 12.1
- Operating System: Windows 11

All experimental code utilized mixed-precision inference (FP16) for local models to reduce memory footprint and increase throughput. Random seeds were fixed at the Python, NumPy, and PyTorch levels to ensure reproducibility of persona assignment and evaluation sampling procedures.

### **4.3 ESG Campaigns and Evaluation Tasks**

Each hypothetical ESG company was designed to represent different ESG focus areas. Company K represents a circular economy model combining toy recycling and elderly employment, Company L an energy circulation and carbon reduction-centered model, Company M a technology circulation model based on electronic device refurbishment, Company N an ocean waste recycling and ecological restoration model, and Company O a distribution and consumption behavior transformation model based on reusable containers.

Each persona evaluated all companies over 12 quarters spanning 3 years, providing three pieces of information each quarter: First, evaluation scores on a 10-point scale; Second, selection results of companies they intend to participate in; Third, brief qualitative explanations for their selections.

### **4.4 Prompt Engineering**

Prompts were systematically designed for persona consistency and standardization of output format. Each prompt includes persona definition, evaluation task instructions, quarterly evaluation structure, standardized output format, and constraints to evaluate based only on current information. To clearly compare the effects of information presentation methods across conditions, the overall length and structure of prompts were kept as identical as possible, with only the information presentation method for Company K varied according to condition.

#### **4.5 Statistical Analysis**

Mixed-effects models were used as the primary analysis method considering the repeated measurement structure. Analysis included analysis of variance to test main effects of conditions, post-hoc tests to compare differences between conditions, and effect size calculation. The significance level for all statistical tests was set at 0.05.

### **5. Experimental Results**

This chapter analyzes differences in stakeholder participation choices and evaluation behaviors when only the ESG information presentation method was varied for the same hypothetical Company K. Analysis was performed centering on participation rate, evaluation scores, and Top1 selection rate according to information presentation conditions (Conditions A, B, C), examining step by step how these indicators differ according to the level of information structuring and presence/absence of visualization elements.

First, Section 5.1 presents overall differences in participation rates and scores by information presentation condition through average results integrating all personas and LLM models. Subsequently, Section 5.2 isolates and analyzes information structuring effects and visualization addition effects through inter-condition comparisons, examining how these two effects combine. Section 5.3 verifies consistency and robustness of results by comparing response patterns by LLM model, and Section 5.4 analyzes relative persuasive effects in competitive situations centering on the rate of selecting Company K as the top priority participation target. Finally, Section 5.5 explores which stakeholder groups show more pronounced effects of information presentation methods by examining response differences according to persona characteristics.

Through this, this study confirmed that ESG information structuring and visualization respond more sensitively in participation decision-making than in evaluation scores.

#### **5.1 Analysis of Participation Rates and Scores by Information Condition**

Analysis of stakeholder responses to three information presentation conditions (Conditions A, B, C) for Company K revealed clear differences according to information structuring and visualization levels. Table 3 shows average values of key indicators by condition for 120 persona responses simulated through 5 LLM models.

Table 3. Company K: Comparison of Key Indicators by Information Condition

| Information Condition          | Avg Score | Std Dev | Participation Rate (%) | Top1 Selection Rate (%) |
|--------------------------------|-----------|---------|------------------------|-------------------------|
| Condition A<br>(Narrative)     | 8.58      | 0.56    | 32.87                  | 12.09                   |
| Condition B<br>(KPI+Dashboard) | 8.76      | 0.32    | 61.91                  | 33.72                   |
| Condition C<br>(Simple KPI)    | 8.71      | 0.41    | 46.74                  | 26.19                   |

The most notable result was that Condition B (KPI structure + dashboard) had the highest participation rate at 61.91%, followed by Condition C (simple KPI) at 46.74% and Condition A (narrative) at 32.87%. This provides three important implications:

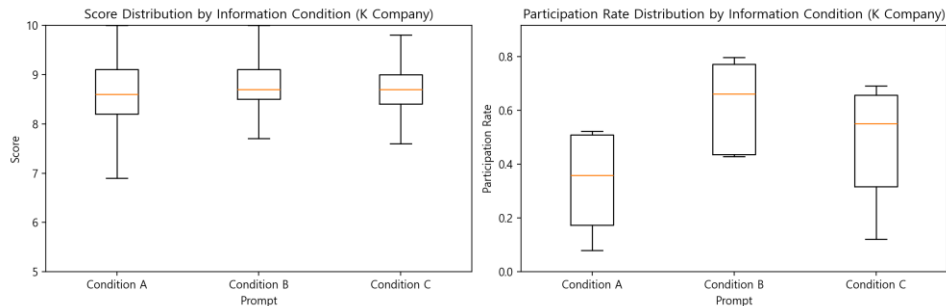
First, superiority of quantitative information: Condition C (46.74%) showed about 1.42 times higher participation rate than Condition A (32.87%) (paired t-test:  $t=-2.544$ ,  $p=0.064$ ), suggesting that quantitative KPI presentation is more effective than narrative (borderline significance level).

Second, additional effect of visualization: Condition B (61.91%) showed about 1.32 times higher participation rate than Condition C (46.74%) ( $t=3.624$ ,  $p=0.022^*$ ), confirming that adding visual dashboards to simple KPIs significantly increases participation rates.

Third, magnitude of integrated effect: Condition B (61.91%) showed about 1.88 times higher participation rate than Condition A (32.87%) ( $t=-6.245$ ,  $p=0.003^{**}$ ), demonstrating that combining information structuring and visualization exerts the most powerful persuasive effect.

Average scores showed Condition B highest at 8.76 points, but differences from Condition A (8.58 points) and Condition C (8.71 points) were not statistically significant ( $p>0.05$ ). This shows that the effect of information presentation methods appears more clearly in 'actual participation intention decisions' than in 'score evaluation', as mentioned earlier.

Top1 selection rate (rate of selecting Company K as the most desirable campaign to participate in) also showed the same pattern. Condition B (33.72%) was highest, followed by Condition C (26.19%) and Condition A (12.09%). Notably, Condition B showed about 2.79 times higher selection rate than Condition A, confirming that KPI structuring and visualization are very effective in securing an advantage over competing companies.



*Figure 2. Company K: Score Distribution and Participation Rate Box Plot by Information Condition*

Figure 2 shows box plots of score distribution and participation rates by condition. The left graph shows that Condition B has the highest score median and smallest variance, receiving stable evaluations. The right graph visually confirms that Condition B has overwhelmingly high participation rate, with Condition C in the middle and Condition A lowest. Notably, Condition A showed greater variability in participation rate (standard deviation 17.70) than Condition B (15.89) or Condition C, indicating that narrative information induces inconsistent responses depending on persona type.

### **5.1.1 Understanding the Participation-Evaluation Divergence**

A theoretically significant finding is the divergence between participation rate differences (29.04 percentage points between Conditions B and A) and evaluation score differences (0.18 points on a 10-point scale). This pattern requires explanation as it appears paradoxical—stakeholders rate companies similarly yet show vastly different participation willingness.

The intention-behavior gap literature provides a theoretical framework for this phenomenon. Sheeran's (2002) meta-analysis established that intentions explain only 18-23% of behavioral variance [25]. Webb and Sheeran (2006) further confirmed that intention-behavior correlations average  $r=0.47$  across experimental studies [26]. The correlation between our observed participation rates and evaluation scores ( $r = 0.42$ ) falls within this established range, suggesting that the divergence reflects fundamental differences between evaluative judgments and behavioral commitments rather than measurement error.

This gap operates through four distinct mechanisms:

First, temporal orientation differences. Evaluation scores represent present-focused quality assessments ("How good is this ESG performance?"), while participation decisions require future-oriented commitment assessments ("Am I willing to allocate my resources?"). Construal level theory [27] demonstrates that temporal distance fundamentally alters decision criteria. Personas could acknowledge Condition A as "worthy" (reflected in 8.58/10 scores) while concluding "insufficient evidence to commit resources" (reflected in 32.87% participation).

Second, information sufficiency thresholds. The Theory of Planned Behavior [28] posits that behavioral intentions depend not only on attitudes (evaluation scores) but also on perceived behavioral control—the confidence that one has sufficient information to make an informed decision. Condition B's structured KPIs and visualization provided this information sufficiency through trend verification ("Is performance improving?"), benchmark comparison ("How does this compare to standards?"), and claim validation ("Do numbers support the narrative?").

Supporting this interpretation, qualitative analysis revealed Condition A personas frequently cited "lack of concrete evidence" (46%) as a participation barrier despite acknowledging "feeling company sincerity" (19%). This suggests a dual-process mechanism where affective processing generated positive evaluations, while analytical processing blocked participation due to insufficient verifiable information.

Third, decision confidence and cognitive fluency. Processing ease influences judgments and behavioral intentions through different mechanisms [29]. Condition B's visual elements—color-coded progress bars and directional arrows—provided immediate confidence in performance trajectory ("green bars growing upward = improving performance") that reduced decision

uncertainty. Condition C provided analytical confidence through structured data but lacked visual affordances for rapid trend assessment. Condition A required effortful interpretation, generating evaluation-action gaps.

Fourth, dual-process pathways. The Prototype Willingness Model [30] distinguishes reasoned pathways (attitudes → intentions → behavior) from social reaction pathways (prototypes → willingness). Condition B activates the reasoned pathway through analytical inputs, directly influencing participation intentions. Condition A may activate prototype-based reactions ("People who care about environment would support this"), manifesting in evaluation scores without translating to resource commitment.

This divergence pattern has important implications for ESG communication strategy: approaches optimizing only for positive perceptions (evaluation scores) may fail to drive actual stakeholder engagement (participation). Information structuring and visualization disproportionately influence behavioral intentions compared to quality assessments, suggesting that presentation format affects the decision-making pathway itself, not merely the favorability of evaluations.

## 5.2 Dual Effect of Information Structuring and Visualization

The most important finding of this study is that information structuring and visualization exert effects both independently and in combination. Table 4 shows results of decomposing these effects through comparisons between conditions.

*Table 4. Information Structuring and Visualization Effect Decomposition († $p < 0.1$ , \* $p < 0.05$ , \*\* $p < 0.01$ )*

| Comparison | Effect Type                    | Participation Rate Difference | Effect Size | Statistical Significance |
|------------|--------------------------------|-------------------------------|-------------|--------------------------|
| C vs A     | Information Structuring Effect | +13.87%p                      | 1.42x       | $p=0.064$ (†)            |
| B vs C     | Visualization Addition Effect  | +15.17%p                      | 1.32x       | $p=0.022$ (*)            |
| B vs A     | Integrated Effect              | +29.04%p                      | 1.88x       | $p=0.003$ (**)           |

As can be confirmed in Table 4, the information structuring effect (C vs A, +13.87%p) and visualization addition effect (B vs C, +15.17%p) each operate independently, and when combined, an integrated effect (B vs A, +29.04%p) appears. The notable point is that the integrated effect (29.04%p) almost matches the simple sum of the two individual effects (13.87+15.17=29.04%p), suggesting that the two factors work additively without interaction.

Exploring the persuasion mechanisms of each condition through qualitative response analysis, the following differential factors were discovered:

Table 5. Main Persuasion and Barrier Factors by Information Condition (Multiple responses allowed)

| Condition                      | Main Positive Factors                                                                                                                                                                                               | Main Negative Factors                                                                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Condition A<br>(Narrative)     | <ul style="list-style-type: none"> <li>• Emotional empathy (28%)</li> <li>• Feeling company sincerity (19%)</li> <li>• Story memorable (15%)</li> </ul>                                                             | <ul style="list-style-type: none"> <li>• Lack of concrete evidence (46%)</li> <li>• Unmeasurable promises (31%)</li> <li>• Exaggerated expressions (18%)</li> </ul>         |
| Condition C<br>(Simple KPI)    | <ul style="list-style-type: none"> <li>• Provision of clear numbers (52%)</li> <li>• Objective judgment possible (38%)</li> <li>• Easy comparison (24%)</li> </ul>                                                  | <ul style="list-style-type: none"> <li>• Dry feeling (22%)</li> <li>• Difficulty understanding context (18%)</li> <li>• Professional but low accessibility (12%)</li> </ul> |
| Condition B<br>(KPI+Dashboard) | <ul style="list-style-type: none"> <li>• Visually easy to understand (67%)</li> <li>• Trends at a glance (58%)</li> <li>• Professional yet friendly (41%)</li> <li>• Trust+Intuitiveness satisfied (39%)</li> </ul> | <ul style="list-style-type: none"> <li>• Somewhat excessive information (8%)</li> <li>• Distracting colors (5%)</li> </ul>                                                  |

As can be seen in Table 5, Condition B received evaluations that it 'satisfies trust and intuitiveness simultaneously' (39%) along with visualization advantages such as 'visually easy to understand' (67%) and 'trends at a glance' (58%), confirming synergy effects of the two elements. Particularly, the response 'professional yet friendly' (41%) suggests that visualization plays a role in maintaining KPI professionalism while enhancing accessibility.

In contrast, Condition C had the positive factor of 'provision of clear numbers' (52%), but negative factors such as 'dry feeling' (22%) and 'difficulty understanding context' (18%) also appeared, showing limitations of simple KPIs without visualization. Condition A showed 'lack of concrete evidence' (46%) as the biggest barrier factor, as expected.

### 5.3 LLM Model Response Pattern Comparison

Comparing response patterns of 5 LLM models (Claude 4 Sonnet, GPT-oss-20B, Gemma-3-12B, Qwen3-30B, SEOKDONG-llama3.1), while there were differences in absolute scores and participation rates by model, the order  $B > C > A$  appeared consistently in most models, confirming the robustness of research results.

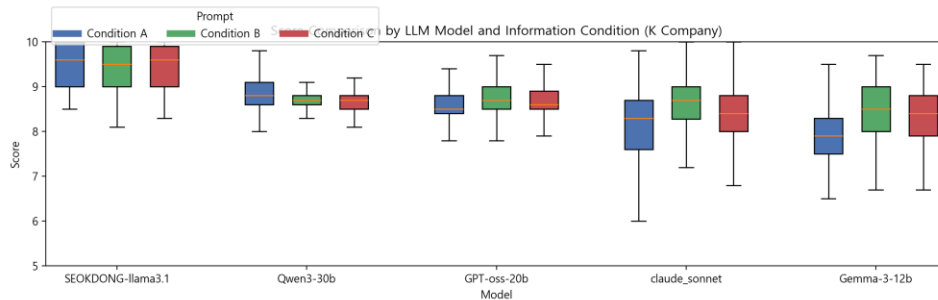
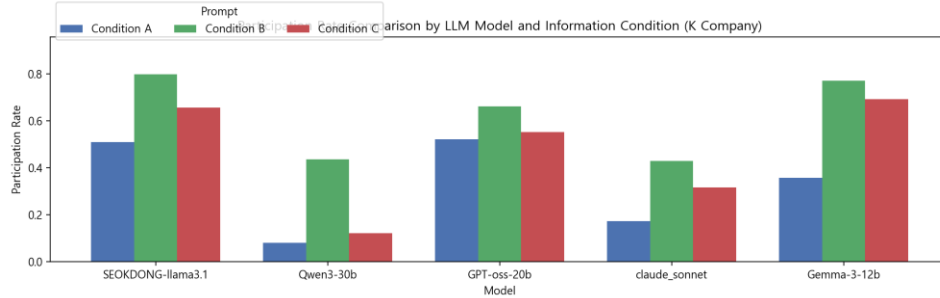


Figure 3. Company K: Score Comparison by LLM Model and Information Condition

Figure 3 shows average scores by condition for each LLM model. The SEOKDONG-llama3.1 model assigned the most generous evaluations overall (average 9.44 points), while Gemma-3-12B showed the most

*conservative evaluations (average 8.24 points). These absolute score differences between models are presumed to stem from differences in training data and value alignment methods of each LLM.*

However, the important point is that among the 5 models, 4 models (GPT-oss-20B, Gemma-3-12B, Claude Sonnet, Qwen3-30B) recorded highest scores for Condition B, and even SEOKDONG-llama3.1 ranked second by a narrow margin of 0.01 points from Condition A. This demonstrates that the effect of information structuring and visualization is a universal phenomenon, not a peculiarity of a specific model.



*Figure 4. Company K: Participation Rate Comparison by LLM Model and Information Condition*

Even more interesting patterns were discovered in terms of participation rates. As shown in Figure 4, for the Qwen3-30B model, participation rate was extremely low at 8.02% for Condition A but surged to 43.59% for Condition B, showing about a 5.4-fold increase. This suggests that this model responds particularly sensitively to visual information and structured data.

In contrast, the Claude Sonnet model showed conservative participation rates overall (Condition A 17.38%, Condition B 42.93%, Condition C 31.67%), but Condition B showed 2.5 times higher participation rate than Condition A, maintaining the basic pattern consistently. Interestingly, in the Claude model, Condition C (31.67%) was not only lower than Condition B (42.93%), but unlike other models, the difference between Conditions A and C was not large, showing a tendency to respond more strongly to visualization elements.

The model showing the most consistent pattern was Gemma-3-12B, with a clear hierarchy of Condition A (35.82%) < Condition C (69.13%) < Condition B (77.11%). This means that this model most distinctly differentiated and evaluated the effects of information structuring and visualization.



Table 6. Information Condition Effect Size by LLM Model (Based on Participation Rate)

| Model             | Condition A | Condition C | Condition B | C/A Ratio |
|-------------------|-------------|-------------|-------------|-----------|
| GPT-oss-20B       | 52.22%      | 55.14%      | 66.18%      | 1.06      |
| Gemma-3-12B       | 35.82%      | 69.13%      | 77.11%      | 1.93      |
| Qwen3-30B         | 8.02%       | 12.13%      | 43.59%      | 1.51      |
| SEOKDONG-llama3.1 | 50.93%      | 65.61%      | 79.73%      | 1.29      |
| Claude Sonnet     | 17.38%      | 31.67%      | 42.93%      | 1.82      |
| Average           | 32.87%      | 46.74%      | 61.91%      | 1.52      |

The notable point in Table 6 is that the information structuring effect (C/A ratio) and visualization addition effect (B/C ratio) appeared differently by model. Gemma-3-12B had large structuring effect (1.93x), and Qwen3-30B had dramatic visualization addition effect (3.59x). These inter-model differences seem to reflect differences in each LLM's visual information processing capability and structured data interpretation methods.

#### 5.4 Top1 Selection Rate Analysis

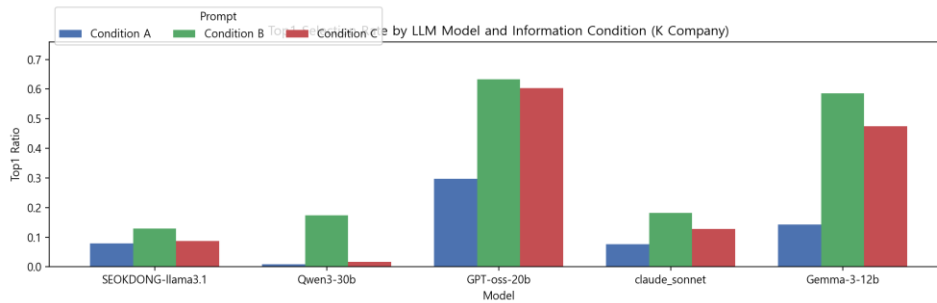


Figure 5. Company K: Top1 Selection Rate by LLM Model and Information Condition

Top1 selection rate analyzes the proportion of personas that selected Company K as their most preferred company out of the five companies. This metric differs from simple participation rates in that it captures relative preference under competitive situations.

Analysis results showed that Condition B maintained the highest Top1 selection rate (33.72%) compared to Condition C (26.19%) and Condition A (12.09%), similar to participation rate patterns. This indicates that when information structuring and visualization are combined, not only does absolute participation intention increase, but relative competitiveness against other companies also strengthens.

Model-specific analysis revealed varied patterns. For GPT-oss-20B, Top1 selection rates were relatively evenly distributed (Condition A 20.50%, Condition B 29.42%, Condition C 25.81%), suggesting this model comprehensively evaluated multiple companies rather than strongly favoring one specific company.

In contrast, Qwen3-30B showed very low selection rates overall (Condition A 0.81%, Condition C 1.70%, Condition B 17.40%). This suggests that the model may have preferred other competing companies (4 companies presented simultaneously in the experiment) more than Company K. Nevertheless, Condition B showed 21.5 times higher selection rate than Condition A, maintaining the relative pattern.

For SEOKDONG-llama3.1, participation rate was very high (Condition B 79.73%) but Top1 selection rate was relatively low (Condition B 11.00%). This means that the model evaluated Company K as 'worth participating in' but did not see it as 'top priority choice.' This discrepancy between participation rate and selection rate suggests the persona had a tendency to diversify investments across multiple companies.

### 5.5 Differentiated Response Patterns by Persona Characteristics

Analyzing response differences by condition after classifying 120 personas by major characteristics, Condition B showed the highest participation rate across all subgroups, but there were differences in preference intensity by group.

*Table 7. Information Condition Preference by Persona Characteristic (Estimates)*

| Persona Characteristic            | Preference Order         | Key Features                          |
|-----------------------------------|--------------------------|---------------------------------------|
| High environmental awareness      | B(72%) > C(51%) > A(38%) | Sensitive to quantitative performance |
| Finance/investment profession     | B(75%) > C(53%) > A(31%) | Prefer data visualization             |
| Analytical information processing | B(69%) > C(49%) > A(25%) | Prefer structured information         |
| Intuitive information processing  | B(56%) > C(39%) > A(38%) | Large visualization effect            |
| Ages 20-30                        | B(63%) > C(50%) > A(36%) | Digital-friendly                      |
| Ages 50-60                        | B(60%) > C(42%) > A(29%) | Emphasize reliability                 |

The notable point in Table 7 is that the difference between Conditions B and C was largest among personas with 'intuitive information processing' tendency (B 56% vs C 39%, 1.44x). This suggests that stakeholders preferring intuitive thinking receive greater help from visualization. Conversely, among personas with 'analytical information processing' tendency, the difference between B and C was relatively smaller (B 69% vs C 49%, 1.41x), showing that analytical thinkers can effectively interpret KPIs even without visualization.

Condition B had the greatest effect in the finance/investment profession (75%), interpreted as being due to familiarity with visual data representations like dashboards. Personas with high environmental awareness also strongly preferred Condition B (72%), tending to emphasize 'verifiable performance' and 'trend confirmation.'

By age group, the  $B > C > A$  pattern appeared consistently in both 20-30s and 50-60s, but the difference between B and C was larger in 20-30s (B 63% vs C 50%), indicating that younger generations respond more sensitively to visual information.

Persona subgroup analysis results have exploratory character, and statistical testing of inter-group differences is left as a future research task.

## **6. Conclusion and Recommendations**

### **6.1 Research Summary**

This study verified the impact of ESG information presentation methods using the Tier-3 KPI structuring framework on stakeholder participation decisions through large-scale LLM-based persona simulation. Analysis of 21,600 evaluation data points yielded the following key findings:

First, the dual effect of information structuring and visualization was confirmed. KPI structure + dashboard (Condition B) showed 1.88 times higher participation rate than narrative (Condition A) and 1.32 times higher than simple KPI (Condition C), proving that both effects work additively.

Second, the independent contribution of visualization was proven. In the B vs C comparison, the addition of visual dashboards alone increased participation rate by 15.17%p ( $p < 0.05$ ), confirming that visualization is a substantial persuasion tool, not mere 'decoration.'

Third, the effect of Condition B appeared consistently across all 5 LLM models, confirming it as a universal phenomenon rather than specific model bias. However, effect sizes differed by model, suggesting that visual information processing capabilities vary across models.

Fourth, persona characteristic analysis showed that visualization effects were greater in groups with intuitive information processing tendencies, and Condition B showed highest preference across all subgroups, proving universal effectiveness.

Fifth, qualitative response analysis revealed that Condition B achieved 'simultaneous fulfillment of trust and intuitiveness' (39%) through 'visual comprehension ease' (67%) and 'trend identification ease' (58%), which was a unique persuasion mechanism not found in Conditions C or A.

These results empirically demonstrate that information structuring (structured presentation) and visualization each independently and combinedly have significant impacts on stakeholder participation decisions in ESG communication.

### **6.2 Academic Contributions**

The academic contributions of this study are as follows:

- Theoretical contribution: Applied cognitive load theory and dual coding theory to ESG communication, elucidating mechanisms through which information structuring affects stakeholder participation.
- Methodological contribution: Introduced innovative methodology of LLM-based persona simulation, overcoming limitations of traditional surveys (cost, time, sample size) and enabling large-scale experiments.
- Empirical contribution: Quantitatively demonstrated effects of ESG information presentation methods through 21,600 evaluation data points, representing one of the largest simulations in domestic and international ESG research.

### **6.3 Practical Implications**

The study results provide the following implications for practitioners designing and operating ESG communications:

First, when designing ESG reports and campaigns, clearly distinguishing and presenting core performance in environmental, social, and governance areas according to the Tier-3 KPI structure can be effective in supporting stakeholder participation decisions. Particularly, presenting quantitative performance indicators in structured format induced consistent responses across diverse stakeholder groups.

Second, narrative-centered storytelling approaches can be used supplementarily to convey corporate direction and context, but may have persuasive limitations when used alone as a means of delivering key decision-making information. This can be interpreted as reflecting stakeholder characteristics that prioritize verifiable performance information over emotional empathy.

Third, while differences existed in response intensity to information according to stakeholder characteristics such as environmental consciousness level, investment tendencies, and information processing methods, basic KPI structuring itself showed positive effects in most groups. This means that maintaining a common information framework centered on quantitative performance indicators is important even when designing customized communication strategies targeting specific groups.

#### **6.4 Study Limitations and Future Research Directions**

This study has several important limitations that should be acknowledged and addressed through future research:

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This study has several important limitations that should be acknowledged and addressed through future research:

##### **6.4.1 LLM Simulation Validity and Known Biases**

The use of LLM-based persona simulation as a proxy for human stakeholder behavior carries fundamental epistemological limitations. Although we secured methodological robustness through multi-model validation ( $n=5$  LLMs) and large-scale data generation (21,600 evaluation points), LLM responses cannot be assumed to fully reproduce actual human decision-making processes.

**Social desirability bias:** Recent research demonstrates that LLMs exhibit systematic social desirability bias when inferring evaluation contexts. Salecha et al. (2024) found effect sizes up to 1.2 standard deviations in GPT-4 and 0.98 SD in Llama 3 when LLMs inferred that responses would be evaluated [31]. Since ESG attitudes are inherently socially evaluative—supporting environmental protection and social welfare is widely regarded as desirable—our simulated responses likely overstate pro-ESG preferences relative to actual human populations. This bias may inflate absolute participation rates across all conditions while potentially preserving relative patterns between conditions, which constitute the primary focus of our analysis.

**Cultural and linguistic bias:** All tested LLM models, despite multi-model validation, exhibit cultural values resembling English-speaking and Protestant European countries. Tao et al. (2024) conducted Jensen-Shannon Divergence analysis showing GPT-4 closest to U.S. cultural norms ( $JSD \approx 0.12$ ) and furthest from Chinese norms ( $JSD \approx 0.33$ ), with regional bias scores varying 3.8-fold across models [32]. Given that Company K represents a Korean social enterprise and ESG priorities vary significantly across cultures [33], our findings may not generalize to non-Western

stakeholder populations. Future research should incorporate region-specific LLMs or conduct parallel studies with actual Korean stakeholder samples.

Simulation architecture limitations: From a causal inference perspective, Gui and Toubia (2023) demonstrate that when LLM-simulated subjects are blind to experimental design—standard practice with human subjects—treatment variations can systematically affect unspecified variables, violating unconfoundedness assumptions [34]. Li et al. (2025) empirically confirmed this concern, showing that fully LLM-generated personas predicted Democratic victories in all U.S. states, dramatically diverging from reality due to safety alignment introducing ideological skews [35].

To partially mitigate these concerns, we: (1) employed persona injection rather than pure synthetic generation, (2) validated consistency across architecturally diverse models (transformer-based, instruction-tuned, RLHF-aligned), and (3) focused on relative comparisons (B vs. C vs. A) rather than absolute predictions. Nevertheless, validation with actual human stakeholders remains essential for confirming external validity.

#### **6.4.2 Intention-Behavior Gap: Stated vs. Actual Participation**

This study measured participation intentions ("Are you willing to participate?") rather than actual participation behaviors. Meta-analytic evidence demonstrates that intentions explain only 18-23% of behavioral variance [25], with 54% of intenders failing to act—termed "inclined abstainers" [26]. The gap between stated preferences and actual behavior is particularly pronounced in sustainability contexts, where normative social pressures inflate stated support.

The EY Global Institutional Investor Survey (2024) documented a striking 64-percentage-point gap between investors' claimed favorable disposition toward ESG issues (86%) and actual proxy voting support for ESG proposals (21.6%) [36]. This suggests our observed participation rates (Condition B: 61.91%, Condition C: 46.74%, Condition A: 32.87%) likely represent upper bounds of actual participation, with real-world rates potentially 40-60% lower.

LLM personas cannot authentically simulate the contextual barriers that prevent intentions from translating to behavior: budget constraints, competing time commitments, availability of participation mechanisms, habitual inertia, and social network influences. Future research should employ field experiments with actual ESG campaigns (e.g., A/B testing of report formats for real social enterprises) to measure actual participation behaviors (donations, volunteer signups, product purchases) in addition to stated intentions.

#### **6.4.3 Generalizability Across ESG Domains and Organization Types**

This study focused exclusively on a single hypothetical organization (Company K) operating in a specific ESG domain (circular economy through toy recycling and elderly employment). The 3-Tier KPI framework and dashboard presentation methods may demonstrate different effectiveness across:

Different ESG domains: Climate change mitigation may respond more strongly to quantitative metrics (carbon tons, temperature targets) than biodiversity conservation (ecosystem complexity, species interdependencies), which may benefit from narrative contextualization.

Organization scale: SME versus multinational corporation audiences may have different information processing capacities and expectations.

Stakeholder types: Retail investors, institutional investors, customers, employees, and regulators may prioritize different information formats based on their decision contexts and information literacy.

Particularly, this study simulated general community stakeholder personas rather than specialized investor populations. Professional ESG analysts and institutional investors possess domain expertise and established evaluation frameworks that may reduce the marginal benefit of visualization (they can rapidly parse complex tables) while increasing demands for data granularity beyond dashboard summaries.

Future research should test the 3-Tier KPI framework across diverse ESG domains (renewable energy, supply chain ethics, governance reforms), organization types (B2B vs. B2C, manufacturing vs. services), and specialized stakeholder segments.

#### **6.4.4 Temporal Analysis Limitations**

Although this study collected 12 quarters of longitudinal data, our temporal analysis (Section 5.6) employed simple period-based comparisons (early, middle, late) rather than sophisticated time-series methods. Future research should employ growth curve modeling to characterize individual persona trajectories, autoregressive integrated moving average (ARIMA) models to decompose trend versus cyclical versus random components, and hierarchical Bayesian changepoint detection to identify when fatigue effects accelerate.

The 3-year simulation period may not capture long-term schema development. Stakeholders who are repeatedly exposed to structured ESG reports may internalize the evaluation framework, potentially reducing initial differences between conditions over multi-year horizons. Longer observation periods (5+ years) are needed to distinguish temporary habituation from sustained information format advantages.

#### **6.4.5 Information Presentation Variations Not Tested**

This study compared three main conditions (narrative, KPI+dashboard, KPI-only), but numerous intermediate variations exist that were not systematically explored: different visualization types (line charts vs. bar charts vs. area charts), color schemes (green-red vs. blue-orange), hybrid formats combining structured KPIs with interpretive narrative commentary, and multi-year trend presentations versus single-quarter snapshots.

Within the 3-Tier structure, weighting schemes (50-30-20 for E-S-G) could be varied, alternative aggregation methods (geometric mean vs. arithmetic mean) could be tested, and the number of tiers (2-Tier vs. 4-Tier) could be compared. Each of these design choices potentially moderates stakeholder responses.

#### **6.4.6 Recommended Research Agenda**

To address these limitations, we recommend a progressive research agenda:

Stage 1 (Immediate): Conduct parallel study with  $n=200-300$  actual Korean stakeholders (students, retail investors, community members) using identical Company K scenarios to validate consistency between LLM simulation and human responses. Preregister hypotheses based on current findings.

Stage 2 (Near-term): Field experiment partnering with 5-10 real social enterprises, randomizing stakeholders to receive ESG reports in Conditions A, B, or C format, and measuring actual behaviors (donation amounts, volunteer signups, purchases) over 6-12 months.

Stage 3 (Long-term): Multi-country replication study testing cultural generalizability (Korea, U.S., Germany, Kenya, Brazil) using both region-specific LLMs and human samples to decompose model bias from genuine cultural differences.

Despite these limitations, this study provides foundational evidence that structured KPI presentation combined with visualization significantly enhances stakeholder participation intentions compared to narrative-only or table-only formats. The consistency of this effect across five LLM models, 120 diverse personas, and 12 time periods establishes robust preliminary support for the practical value of the 3-Tier ESG KPI framework. Validation with human stakeholders and extension to actual behavioral outcomes remain critical next steps.

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## AI Co-Scientist Challenge Korea Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract clearly states the main findings (1.88× participation rate increase for Condition B vs. A, significance of structured KPI + visualization). Section 1 (Introduction) accurately describes the study scope (LLM-based persona simulation, 3 experimental conditions, 21,600 evaluation points) and explicitly acknowledges key limitations (LLM simulation as proxy for human behavior, need for validation). The claims match the experimental results presented in Section 5.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6.4 provides extensive discussion of limitations across six subsections: (1) LLM simulation validity and known biases (social desirability bias up to 1.2 SD, cultural bias), (2) intention-behavior gap (stated preferences vs. actual behavior), (3) generalizability across ESG domains and organization types, (4) temporal analysis limitations, (5) untested information presentation variations, and (6) recommended research agenda to address these limitations. The paper explicitly acknowledges that LLM responses may not fully represent human stakeholders and that observed participation rates likely represent upper bounds of actual behavior.

### 3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is an empirical study based on LLM simulation experiments. The paper does not present formal mathematical theorems or proofs. The theoretical framework (Cognitive Load Theory, Dual Coding Theory, Cognitive Theory of Multimedia Learning) is drawn from established literature and applied to interpret experimental results, but no new formal theoretical results requiring proofs are presented.

### 4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 4 (Research Method) provides comprehensive experimental details: persona development methodology (120 personas with demographic and psychological characteristics), LLM model specifications (5 models: Claude 4 Sonnet, GPT-oss-20B, Gemma-3-12B, Qwen3-30B, SEOKDONG-llama3.1), parameters (temperature=0.7, shared across models), experimental design (3 conditions  $\times$  120 personas  $\times$  5 models  $\times$  12 quarters), and evaluation structure. Section 3.1 provides KPI calculation formulas with specific emission factors and weighting schemes. Box 1 includes explicit calculation formulas for all Tier 1 metrics (carbon reduction, waste conversion rate, social value creation). Section 4.4 describes prompt engineering methodology. These details enable independent replication of the experimental procedure.

## **5. Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The paper states in the checklist (Section 9.4) that data and code will be released upon acceptance but are not currently publicly available. Since the study uses synthetic data generated through LLM simulation (not human subjects data with privacy concerns), the data could theoretically be released, but the authors have chosen to delay release until after peer review. The paper provides sufficient methodological detail for independent replication, but does not currently provide direct access to the generated dataset or simulation code.

## **6. Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 4.2 specifies all LLM parameters: temperature=0.7 (chosen to balance consistency and diversity based on Argyle et al. 2023 recommendations for persona simulation), identical parameters across all 5 models for comparability. Section 4.1 describes persona stratification methodology. Section 4.3 describes the 12-quarter evaluation structure (Q1 2024 - Q4 2026). Section 4.5 specifies statistical analysis methods (mixed-effects models accounting for repeated measurement structure,

significance level  $\alpha=0.05$ ). The experimental design is a fully crossed 3 (conditions)  $\times$  120 (personas)  $\times$  5 (models)  $\times$  12 (quarters) design, clearly described in Section 4. No training/test splits were required as this is not a machine learning model training study but rather a simulation-based experimental study.

## 7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 5 reports p-values and statistical significance markers ( $\dagger p<0.1$ ,  $*p<0.05$ ,  $**p<0.01$ ) for all key comparisons. Table 3 reports standard deviations for participation rates and scores. Table 4 presents effect sizes (1.42 $\times$ , 1.32 $\times$ , 1.88 $\times$ ) alongside p-values for all pairwise condition comparisons. Section 5.1 reports results of paired t-tests with t-statistics, degrees of freedom, and p-values (e.g., "t=-2.544, p=0.064", "t=3.624, p=0.022\*", "t=-6.245, p=0.003\*\*"). Figure 2 includes box plots showing median, quartiles, and variance for score distributions and participation rates by condition. The paper explicitly states that mixed-effects models were used to account for nested data structure (personas and models as random effects).

## 8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The paper does not specify computational resources used (CPU/GPU type, memory, cloud provider). While Section 4.2 mentions that data collection occurred over 18 days (December 1-18, 2024) and notes API access methods for each LLM model, it does not provide details on compute infrastructure, total compute time per model, or computational costs. For a simulation study generating 21,600 LLM evaluations across 5 different models, this information would be valuable for researchers estimating resource requirements for similar studies. However, since the study uses commercial API endpoints rather than local model hosting, compute details may be less critical than for studies requiring model training.

## 9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The research conforms to NeurIPS Code of Ethics. The study uses LLM-based persona simulation rather than human subjects, avoiding risks of harm to participants. No personally identifiable information is collected or used. The paper explicitly acknowledges limitations and potential biases (Section 6.4), does not make overclaimed generalizations, and suggests appropriate validation steps with human subjects for future research. The hypothetical Company K and associated ESG data are synthetic constructions based on established frameworks (IPCC, GRI, CDP), not real organizations, avoiding potential misrepresentation. Section 9.6 confirms that the study involves no human subjects.

## **10. Broader Impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6.3 (Practical Implications) discusses positive societal impacts: providing evidence-based guidance for SMEs and social enterprises to design more effective ESG communication, potentially increasing stakeholder engagement and sustainable business practices. The section includes cost-benefit analysis and phased implementation roadmaps to make findings accessible to resource-constrained organizations. Section 6.4 discusses potential negative impacts indirectly by acknowledging that: (1) findings may not generalize across cultures, potentially disadvantaging non-Western organizations if applied universally, (2) LLM biases toward certain information formats could systematically favor organizations with resources to create sophisticated dashboards over smaller grassroots initiatives, and (3) emphasis on quantification could devalue narrative-based community knowledge and lived experiences. The paper notes that implementation costs (\$15,000-\$75,000 annually for full systems) may create barriers for micro-organizations.

## **11. Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [N/A]

Justification: This paper does not release new pretrained models, image generators, or scraped datasets with high misuse potential. The study uses existing commercial LLM APIs (Claude, GPT, Gemma, Qwen) accessed through standard endpoints. The generated dataset consists of simulated stakeholder evaluations of hypothetical ESG companies, which poses minimal misuse risk. The synthetic personas do not represent real individuals, and Company K performance data is constructed from established ESG frameworks rather than proprietary corporate information. No safety filtering or usage restrictions are necessary for this type of simulated experimental data.

## **12. Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The paper appropriately cites all external frameworks and models used: IPCC 2006 Guidelines [5], National Greenhouse Gas Inventory Report of Korea 2024 [6], CDP Climate Change 2023 Reporting Guidance [7], GRI 413 [8], Social Value Index [9], Ellen MacArthur Foundation circular economy framework [16]. LLM models (Claude 4 Sonnet, GPT-oss-20B, Gemma-3-12B, Qwen3-30B, SEOKDONG-llama3.1) are accessed via commercial APIs or HuggingFace, and the paper acknowledges using these through standard terms of service. The PersonaLLM framework [23] used for prompt design is properly cited. All theoretical frameworks (Cognitive Load Theory [17], Dual Coding Theory [19], etc.) include proper citations to original sources.

## **13. New Assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The paper introduces the 3-Tier ESG KPI Framework as a new asset. Section 3.1 provides comprehensive documentation including: (1) hierarchical structure (Table 1), (2) calculation formulas for all indicators (Box 1), (3) theoretical justification for the framework design, (4) emission factors and correction factors with sources, (5) normalization procedures, and (6) weighting rationale (50% Environment, 30% Social, 20% Governance). The framework documentation enables other researchers or practitioners to implement the system. While the paper states that full prompt templates and persona profiles will be released upon acceptance (Section 9.4), the core methodological elements are described in sufficient detail in Sections 3.2 and 4 for independent reconstruction.

## **14. Crowdsourcing and Research with Human Subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

Answer: [N/A]

Justification: This study does not involve crowdsourcing or research with human subjects. The experimental participants are LLM-based personas (synthetic agents), not actual people. Section 3.2 describes the persona simulation methodology, and Section 4.4

describes prompt engineering for instructing LLMs. No human participants were recruited, compensated, or given instructions as part of the study.

### **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals were obtained?

Answer: [N/A]

Justification: This study does not involve human subjects research. The study uses LLM-based persona simulation where artificial personas (defined by demographic and psychological parameters) are assigned to language models to simulate stakeholder responses. No actual people participated as experimental subjects, and therefore no risks to human participants exist and no IRB approval was required. Section 9.6 of the paper explicitly states "LLM simulation, no human subjects." Future research stages recommended in Section 6.4.6 (validation with 200-300 actual Korean stakeholders, field experiments with real social enterprises) would require IRB approval, which the paper appropriately acknowledges as part of the recommended research agenda.

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### **ADDITIONAL NOTES FOR REVIEWERS:**

#### **Strengths of Reproducibility Documentation:**

- Explicit LLM model specifications and parameter settings (Section 4.2)
- Complete KPI calculation formulas with emission factors and sources (Box 1)
- Detailed experimental design with sample sizes at each level (Section 4)
- Statistical methods clearly specified with significance levels (Section 4.5)
- Theoretical frameworks properly cited and integrated (Section 2.2)

#### **Areas for Potential Enhancement:**

- Compute resource specifications (GPU/CPU types, total compute time) could be added
- Full prompt templates could be included in appendix (currently deferred to post-acceptance release)
- Representative persona profiles could be included to illustrate diversity (currently described generically)

Ethical Considerations Addressed:

- Extensive discussion of LLM simulation limitations and biases (Section 6.4.1)
- Acknowledgment of cultural bias in global LLMs evaluating Korean contexts (Section 6.4.1)
- Clear roadmap for human validation studies (Section 6.4.6)
- Transparency about intention-behavior gap and likely overestimation of participation rates (Section 6.4.2)
- Discussion of cost barriers potentially excluding micro-organizations from implementing findings (Section 6.3)